SOFTWARE ENGINEERING

Requirements Engineering & UML Use-Case Modelling

Blood Bank Inventory & Emergency Donor Matcher

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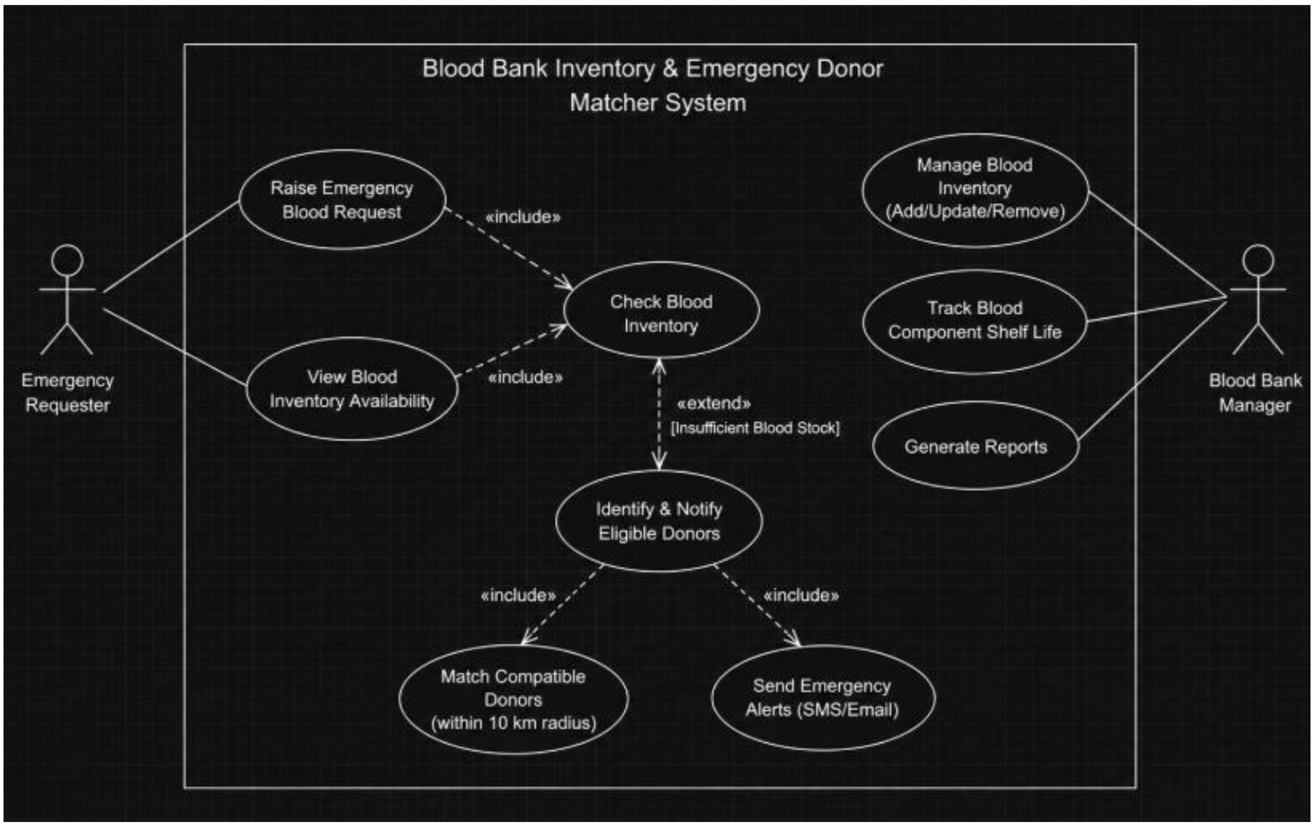
1. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow an Emergency Requester to submit an emergency blood request containing the required blood group, component, quantity, and urgency.	High	Pass: A complete request is recorded and assigned an emergency status. Fail: An incomplete request is accepted.	Enables the system to receive and process critical blood requirements.
FR-002	Functional	The system shall cross-reference an emergency blood request against the real-time blood bank inventory and show whether the required blood component is available.	High	Pass: Current availability is displayed for the requested blood component. Fail: The system uses outdated inventory data.	Determines whether existing stock can satisfy an emergency request.
FR-003	Functional	The system shall identify eligible and compatible donors for the requested blood requirement using blood-group compatibility and the donor's location.	High	Pass: Only compatible and eligible donors are selected. Fail: An incompatible donor is selected.	Provides safe and targeted donor matching during shortages.
FR-004	Functional	The system shall broadcast emergency SMS alerts to compatible eligible donors within a 10 km radius when the required blood stock is critically insufficient.	High	Pass: Matching donors within 10 km receive the alert within 30 seconds of the emergency flag. Fail: Incompatible donors are alerted.	Supports rapid donor mobilization during critical shortages.
FR-005	Functional	The system shall allow the Blood Bank Manager to manage blood inventory, including recording allocations and tracking blood component shelf life.	High	Pass: Inventory quantities and shelf-life information are updated correctly after an allocation or inventory change. Fail: The inventory record remains inconsistent.	Keeps stock information current and helps prevent use of expired components.
NFR-001	Non-Functional – Performance & Security	The inventory ledger shall maintain strict transactional consistency and prevent simultaneous allocation of the same blood bag.	High	Pass: Concurrent allocation tests show that a blood bag cannot be allocated to more than one request. Fail: Duplicate allocation occurs.	Prevents inventory corruption and unsafe duplicate allocation.
NFR-002	Non-Functional – Performance	The system shall process an	High	Pass: Performance testing confirms donor	Ensures the system is responsive enough for

		emergency request and initiate donor notification within 30 seconds under the expected peak emergency workload.		notification is initiated within 30 seconds under simulated peak load. Fail: The target response time is exceeded.	time-critical emergencies.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: Process Emergency Blood Request

Primary Actor: Emergency Requester

Supporting Actor: Blood Bank Manager

Goal: Obtain the required blood component quickly by checking available inventory and, when necessary, matching and alerting compatible eligible donors.

Preconditions: 1. The blood bank system is operational.
2. Current blood inventory is available to the system.
3. Donor eligibility, compatibility, and location information is available for matching.

Postconditions: 1. The emergency request is recorded with its status.
2. The inventory is updated if a blood unit is allocated.
3. If stock is insufficient, compatible eligible donors within the defined radius are identified and emergency alerts are initiated.

Main Success Scenario

1. Emergency Requester submits an emergency blood request with the required blood group, component, quantity, and urgency.
2. System validates the request details and records the emergency request.
3. System checks the real-time blood bank inventory for the requested blood component.
4. System confirms that the required stock is available.
5. Blood Bank Manager reviews/authorizes the allocation.
6. System allocates the required blood unit(s) and updates the inventory ledger.
7. System records the completed allocation and the request is marked as fulfilled.

Alternate Flow – Insufficient Blood Stock

A1. At Step 4, the system determines that the required blood stock is insufficient.

A2. System activates donor matching using blood-group compatibility, eligibility, and location.

A3. System identifies compatible eligible donors within a 10 km radius.

A4. System initiates emergency SMS alerts to the matched donors.

A5. The emergency request remains open until sufficient blood is allocated or the request is otherwise resolved.